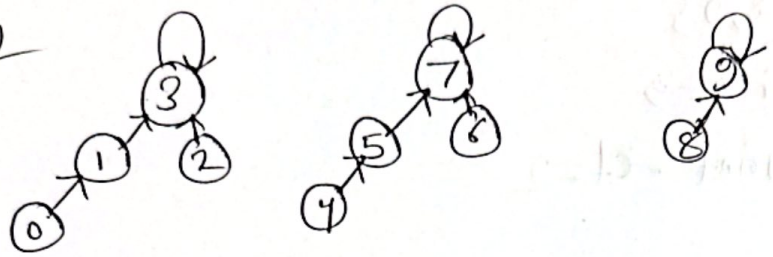
Spring 2026

1)

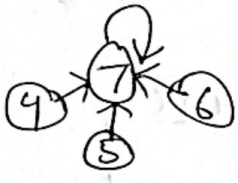
a)

(i)



(ii)

Find (4) → 7

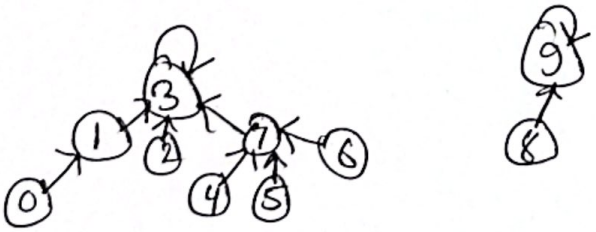


Index	0	1	2	3	4	5	6	7	8	9
Parent	1	3	3	3	7	7	7	7	9	9

(iii)

Find (2) → 3

Find (6) → 7



Index	0	1	2	3	4	5	6	7	8	9
Parent	1	3	3	3	7	7	7	3	9	9
Rank	0	0	0	3	0	0	0	2	0	1

(iv)

Set 1 = {0, 1, 2, 3, 4, 5, 6, 7}

Representative = 3

Set 2 = {8, 9}

Representative = 9

Total disjoint set = 2

(b)

(i) Since Rank[A] > Rank[B]

So, A becomes the parent of B

(ii) since Rank[A] = Rank[C]

So, Rank will increase by 1

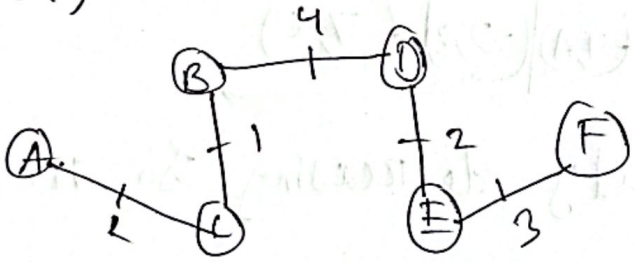
Rank k = 3 + 1 = 4

(2)

(a) Using prims Algorithm;

	A	B	C	D	E	F
Initially	0	∞	∞	∞	∞	∞
A (0,-)		(3,A)	(2A)	∞	∞	∞
C (2,A)		(1,C)		(5,C)	(6,C)	∞
B (1,C)				(4,B)		∞
D (4,B)					(3,D)	(7,D)
E (3,D)						(5,E)
F (5,E)						

Final MST,



(b) Yes, A connected weighted graph can have Multiple MST's when different choices of equal weights edges produce different trees.

(3)

(a) No, the graph has a negative weighted cycle

$A \to B \to C \to A$

Cycle weight : $1 + 2 + (-5) = -2$

(b) Bellman-ford Algorithm will help me to find the shortest Path &:-

	A	B	C	D
Initially	0	∞	∞	∞
1	(-2,C)	(2,A)	(-3,B)	(-1,C)
2	(-4,C)	(0,A)	(-5,B)	(-3,C)
3	(-6,C)	(-2,A)	(-7,B)	(-5,C)
4	(-8,C)	(-4,A)	(-9,B)	(-7,C)

Path's:
A → B
B → C
B → D
C → A
C → D

It is continuously decreasing so, no shortest path exists

(4)
(i) $h'(46) = (2 \times 46 + 5) \bmod 11$
$= 97 \bmod 11$
$= 9$
$h(46,0) = (h'(48) + 3 * 0^2) \bmod 11$
$= (9+0) \bmod 11$
$= 9$

Hash table after inserting 46,

0	1	2	3	4	5	6	7	8	9	10
14	Nil	Nil	Nil	60	33	Nil	84	Nil	46	Nil

(ii)

$h'(82) = (2 \times 82 + 5) \bmod 11$

$= 4$

$h(82, 0) = (h'(82) + 3 \times 0^2) \bmod 11$

$= (4 + 3 \times 0) \bmod 11$

$= 4 \Rightarrow$ collision.

$h(82, 1) = (4 + 3 \times 1^2) \bmod 11$

$= 7 \rightarrow$ collision

$h(82, 2) = (4 + 3 \times 2^2) \bmod 11$

$= 5 \Rightarrow$ collision

$h(82, 3) = (3 \times 3^2 + 4) \bmod 11$

$= 9 \rightarrow$ collision

$h(82, 4) = (4 + 3 \times 4^2) \bmod 11$

$= 8 \rightarrow$ insert.

0	1	2	3	4	5	6	7	8	9	10
14	Nil	Nil	Nil	60	33	Nil	34	82	46	Nil

(iii)

34 is stored at index 7

$\therefore T[7] =$ Nil

0	1	2	3	4	5	6	7	8	9	10
14	Nil	Nil	Nil	60	33	Nil	Nil	82	46	Nil

(iv)

Search 82 ⇒

$h(82,0) = 4$

$h(82,1) = 7 \rightarrow$ NIL ⇒ Stop

82 Not Found, though it exists at index 8

To prevent this we use Deleted marker (Tombstone) instead of Nil

(v) Merits: Quadratic Probing removes primary clustering

Demerits: It suffers from secondary clustering because keys have the same initial hash value follow the same probe sequence.

(5)

(a) $\#(P) = (20+9+20+12+5)$

$= 66$

(b)

$\#(T) =$ SAVESTITLE

Index	Window	Rolling Hash Func	Remark
0	SAVES	19+1+22+5+19=66	Spurious Hit
1	AVEST	66−19+20=67	No Match
2	VESTI	67−1+9=75	No Match
3	ESTIT	75−22+20=73	No match
4	STITL	73−5+12=80	No match
5	TITLE	80−19+5=66	Match

(c) Worst case Time complexity : $O(nm)$

Frequent spurious hit ~~met~~ needs character by character comparisons.